EUROPEAN STANDARD **EN 18219**

NORME EUROPÉENNE

EUROPÄISCHE NORM May 2026

ICS 35.240.63

English version

Digital product passport - Unique identifiers

Passeport numérique des produits - Identifiants uniques

Digitaler Produktpass - Eindeutige Kennungen

This European Standard was approved by CEN on 3 May 2026.

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Contents

Page

European foreword 4
Introduction 5
1 Scope 7
2 Normative references 7
3 Terms, definitions and abbreviations 8
3.1 Terms and definitions 9
3.2 Abbreviations 11
4 General principles, requirements and guidelines 12
4.1 Global uniqueness 12
4.1.1 Principle 12
4.1.2 Requirements 12
4.2 Persistence 13
4.2.1 Principle 13
4.2.2 Requirements 13
4.3 Syntax 13
4.3.1 Principle 13
4.3.2 Requirements 13
4.3.3 Guidelines 14
4.4 Granularity 14
4.4.1 Principle 14
4.4.2 Requirements 14
4.4.3 Guidelines 14
4.5 Interoperability 14
4.5.1 Principle 14
4.5.2 Requirements 15
4.5.3 Guidelines 15
4.6 Openness 15
4.6.1 Principle 15
4.6.2 Requirements 15
5 ID schemes for products 16
5.1 General 16
5.2 ID scheme 1: Web enabled, structured path and query ID for products 16
5.2.1 Description 16
5.2.2 Requirements 16
5.3 ID scheme 2: Identification Link (IL) string for products 17
5.3.1 Description 17
5.3.2 Requirements 17
5.3.3 Guidelines 17
5.4 ID scheme 3: Decentralized identifiers (DID) for products 17
5.4.1 Description 17
5.4.2 Requirements 18
5.4.3 Guidelines 18
5.5 ID scheme 4: Identification for products and product groups 18
5.5.1 Description 18
5.5.2 Requirements 18
5.5.3 Guidelines 19
5.6 ID scheme 5: Digital Object Identifier (DOI) for products 19
5.6.1 Description 19

5.6.2 Requirements 19
6 ID schemes for economic operators and facilities 19
6.1 General 19
6.2 ID scheme 6: Economic operator identification using ISO/IEC 6523 20
6.2.1 Description 20
6.2.2 Requirements 20
6.2.3 Guidelines 20
6.3 ID scheme 7: Economic operator and facility identification per ISO/IEC 15418 20
6.3.1 Description 20
6.3.2 Requirements 21
6.4 ID scheme 8: Economic operator identification in data systems using Decentralized identifiers (DID) 21
6.4.1 Description 21
6.4.2 Requirements 21
6.4.3 Guidelines 21
6.5 ID scheme 9: Economic operator identification used in data systems per Digital Object Identifiers (DOI) 22
6.5.1 Description 22
6.5.2 Requirements 22
Annex A (informative) Selecting identification granularity levels for products 23
Annex B (informative) Overview of ID schemes for products 25
Annex C (informative) Overview of ID schemes for economic operators and facilities 47
Annex ZA (informative) Relationship between this European Standard and the essential requirements of (EU) 2024/1781 aimed to be covered 56
Bibliography 58

European foreword

This document (EN 18219:2026) has been prepared by Technical Committee CEN/CLC/JTC 24 “Digital product passport – Framework and system”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2026, and conflicting national standards shall be withdrawn at the latest by November 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

According to the CEN-CENELEC Internal Regulations, the national standards organisations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom.

Introduction

0.1 Background

A digital product passport (DPP) is a key enabling mechanism to make product information traceable and accessible across value chains – supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions and to support a circular economy. The implementation of digital product passports will be carried out progressively. Sector-specific initiatives will determine the precise DPP content and requirements for individual product groups.

To support the implementation of DPPs, 8 standards have been developed so far:

— EN 18219:2026 – Digital product passport – Unique identifiers (this document)

— EN 18220:2026 – Digital product passport – Data carriers

— EN 18216:2026 – Digital product passport – Data exchange protocols

— EN 18222: 2026 – Digital Product Passport – Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability

— EN 18223:2026 – Digital Product Passport – System interoperability

— EN 18221:2026 – Digital product passport – data storage, archiving, and data persistence

— EN 18239:—[1] – Digital Product Passport – access rights management, information system security, and business confidentiality

— EN 18246:—[2] – Digital Product Passport – Data authentication, reliability and integrity

0.2 Overview

This document provides the foundation for the identification of products, economic operators, and facilities to be used for digital product passports.

Clause 4 sets out the principles, requirements, and guidelines that apply across all identification use cases. These are structured around six key areas: global uniqueness, persistence, syntax, granularity, interoperability, and openness.

— *Principles:* Provide the intent and purpose for each area.

— *Requirements:* Define what shall be fulfilled when implementing identifiers, serving as the foundation for assessment and compliance.

— *Guidelines:* Offer optional recommendations that can support or enhance implementation but are not mandatory.

Clause 5 specifies the permitted identification schemes for creating a unique product identifier. Five different schemes are described, each of which may be used to meet the requirements established in Clause 4, provided that both the rules of the chosen scheme and the requirements of Clause 4 are fully satisfied.

Clause 6 specifies the permitted identification schemes for creating a unique economic operator identifier and unique facility identifier. Four schemes are described, each of which may be used

[1] Under preparation. Stage at the time of publication: prEN 18239:2025.
[2] Under preparation. Stage at the time of publication: prEN 18246:2025.

to meet the requirements established in Clause 4, provided that both the rules of the chosen scheme and the requirements of Clause 4 are fully satisfied.

In addition, the document includes informative annexes to support practical understanding and implementation:

— *Annex A*: Selecting identification granularity levels for products;

— *Annex B*: Overview of identification schemes for products;

— *Annex C*: Overview of identification schemes for economic operators and facilities;

— *Annex ZA:* Relationship between this European Standard and the essential requirements of 2024/1781.

1 Scope

This document defines the principles and specifies the requirements and guidelines for unique product identifiers, unique economic operator identifiers, and unique facility identifiers used in digital product passports. It covers the following areas:

a) global uniqueness;

b) persistence;

c) syntax;

d) granularity;

e) interoperability;

f) openness.

This document accommodates unique product identifiers at three granularity levels of specificity: model, batch, or individual item, to support various operational needs.

This document describes identification (ID) schemes that use issuing agencies, self-issuing systems, or a combination of both.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 646:1991, *Information technology — ISO 7-bit coded character set for information interchange*

ISO/IEC 6523-1:2023, *Information technology — Structure for the identification of organizations and organization parts — Part 1: Identification of organization identification schemes*

ISO/IEC 6523-2:2025, *Information technology — Structure for the identification of organizations and organization parts — Part 2: Registration of organization identification schemes*

ISO/IEC 10646:2020, *Information technology — Universal coded character set (UCS)*

ISO/IEC 15418:2016, *Information technology — Automatic identification and data capture techniques — GS1 Application Identifiers and ASC MH10 Data Identifiers and maintenance*

ISO/IEC 15434:2025, *Information technology — Automatic identification and data capture techniques — Syntax for high-capacity ADC media*

ISO/IEC 15459-2:2015, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 2: Registration procedures*

ISO/IEC 15459-3:2014, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 3: Common rules*

ISO/IEC 15459-4:2014, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 4: Individual products and product packages*

ISO/IEC 15459-6:2014, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 6: Groupings*

ISO/IEC 15961-1:2021, *Information technology — Data protocol for radio frequency identification (RFID) for item management — Part 1: Application interface*

ISO/IEC 17360:2023, *Automatic identification and data capture techniques — Supply chain applications of RFID — Product tagging, product packaging, transport units, returnable transport units and returnable packaging items*

ISO/IEC 18975:2024, *Information technology — Automatic identification and data capture techniques — Encoding and resolving identifiers over HTTP*

ISO/IEC 19762:2025, *Information technology — Automatic identification and data capture (AIDC) techniques — Vocabulary*

ISO 26324:2025, *Information and documentation — Digital object identifier system*

EN 18220, *Digital product passport - Data carriers*

EN IEC 61406-1:2022, *Identification Link - Part 1: General requirements*

EN IEC 61406-2:2024, *Identification link - Part 2: Types/models, lots/batches, items and characteristics*

DIN 16598:2022, *Syntax keyboard and Web compatible encoding of data elements in machine readable symbols applied with ASC Data Identifiers*

GS1, Digital Link Standard: Syntax U.R.I. Version 1.6.0:2022, https://www.gs1.org/standards/gs1-digital-link

GS1. EPC Tag Data Standard (TDS), Version 2.3:2025, https://www.gs1.org/standards/epc-tag-data-standard

GS1, General Specifications Standard, Version 25.0:2025, https://ref.gs1.org/standards/genspecs/

IETF. RFC 3986:2005, https://www.rfc-editor.org/rfc/rfc3986

W3C, Decentralized Identifier Resolution (DID Resolution) v0.3:2025, https://www.w3.org/TR/did-resolution/

W3C, Decentralized Identifiers (DIDs) v1.0:2022, https://www.w3.org/TR/did-1.0/

W3C, Verifiable Credentials Data Model v2:2025, https://www.w3.org/TR/vc-data-model/

3 Terms, definitions and abbreviations

For the purposes of this document, the terms and definitions given in ISO/IEC 19762:2025 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1 Terms and definitions

3.1.1
batch
subset of a *model* (3.1.11) that is grouped by an economic operator based on common characteristics

3.1.2
consumer
individual member of the general public purchasing or using goods, property or services for private purposes

[SOURCE: EN ISO 14025:2010, 3.16]

3.1.3
digital product passport
DPP
digital record of product characteristics throughout its life cycle that is accessible via electronic means through a data carrier

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability.

3.1.4
domain
specified area of knowledge, application, or activity within which a unique identifier is used

3.1.5
economic operator
manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider that places the product on the market

3.1.6
identification scheme
ID scheme
system for allocating identifiers to objects

[SOURCE: ISO/IEC 6523-1:2023, 3.6, modified — "registered" removed]

3.1.7
interoperability
ability of two or more systems or applications to exchange information and to mutually use the information that has been exchanged

[SOURCE: ISO/IEC 22123-1:2023, 3.6.1]

3.1.8
issuing agency
organisation entrusted by a Registration Authority (RA) to manage unique identification schemes

3.1.9
item
single unit of a *model* (3.1.11)

3.1.10
life cycle
consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal

[SOURCE: ISO 14040:2006, 3.1]

3.1.11
model
version of a product of which all units share the same technical characteristics and the same model identifier

3.1.12
object
entity to which specified requirements apply

EXAMPLE Product, process, service, system, installation, project, data, design, material, claim, person, body or organization, or any combination thereof.

[SOURCE: ISO/IEC 17000:2020, 4.2, modified — the synonym "object of conformity assessment" has been deleted and Note 1 to entry has been deleted]

3.1.13
persistence
process to keep a unique identifier associated with an identified object throughout its life cycle

3.1.14
placed on the market
first making available of a product

3.1.15
preservation
activity of ensuring that a unique identifier and its denotation remain stable, valid, and retrievable for as long as required

3.1.16
product
any physical good that is *placed on the market* (3.1.14) or *put into service* (3.1.17)

3.1.17
put into service
first use, for its intended purpose, of a product

3.1.18
resolution
process in which an identifier is translated or mapped to another identifier, associated information or metadata about the object it represents

3.1.19
resolver
system or service that performs *resolution* (3.1.18)

3.1.20
self-issuing system
decentralized identification scheme

system or mechanism that an organization uses to generate and assign unique identifiers to its objects without the intervention or oversight of an external authority

3.1.21
state-of-the-art
developed stage of technical capability at a given time as regards products, processes and services, based on the relevant consolidated findings of science, technology and experience

[SOURCE: ISO/IEC Guide 2:2004, 1.4]

3.1.22
unique economic operator identifier
unique string of characters for the identification of an actors involved in a product's value chain

3.1.23
unique facility identifier
unique string of characters for the identification of location or building involved in a product's value chain or used by actors involved in a product's value chain

3.1.24
unique identifier
string of characters that is required to be unique among all identifiers used for all objects for a specific purpose

Note 1 to entry: A unique identifier refers to *unique product identifier* (3.1.25), *unique economic operator identifier* (3.1.22), and *unique facility identifier* (3.1.23).

[SOURCE: ISO 29404:2015, 3.26, modified — Note 1 to entry removed and a new Note 1 to entry added]

3.1.25
unique product identifier
unique string of characters for the identification of a product that also enables a web link to the digital product passport

3.2 Abbreviations

For the purposes of this document, the following abbreviations apply.

Abbreviation	Meaning
AIDC	Automatic identification and data capture
AI	Application Identifier (GS1)
ASC	Accredited Standards Committee
DI	Data Identifier (ASC MH10.8.2)
DID	Decentralized Identifier
DOI	Digital Object Identifier
DPP	digital product passport
EPC	Electronic Product Code
EFTA	European Free Trade Association

ETSI	European Telecommunications Standards Institute
GLN	Global Location Number
GLEIF	Global Legal Entity Identifier Foundation
GTIN	Global Trade Item Number
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IAC	Issuing Agency Code
ICD	International Code Designator
ID	Identifier / Identification
IEC	International Electrotechnical Commission
IL	Identification Link
ISO	International Organization for Standardization
LEI	Legal Entity Identifier
MH10	ANSI-Accredited Committee MH10 on Material Handling
RFID	Radio-frequency identification
SGTIN	Serialized Global Trade Item Number
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
VC	Verifiable Credential
W3C	World Wide Web Consortium

4 General principles, requirements and guidelines

4.1 Global uniqueness

4.1.1 Principle

The principle of global uniqueness ensures that each identifier is distinct, at least exclusive within its domain of use, and non-coexisting, enabling clear object differentiation and thus reducing the risk of misidentification. Products receive at least one globally unique product identifier at one of the three levels of granularity (model, batch, or individual item), and when not yet available, the economic operator or facility is assigned a globally unique economic operator identifier or a globally unique facility identifier. The uniqueness is ensured by applying mechanisms, specifications and rules as described in one of the identification schemes in this document when placing products on the market or putting them into service.

4.1.2 Requirements

A unique identifier shall be globally unique, by ensuring:

1) *No reassignment:* The same unique identifier shall not be used to identify different objects.

2) *Distinct:* A unique identifier shall be globally unique within its domain and unambiguous across all potential contexts in which it may be used.

3) *Non-coexistence:* A unique identifier, once issued, shall not coexist with or be reassigned to another object.

4) *Cross domain identification:* To ensure uniqueness across domains, the ID scheme for the unique identifier specified in this document shall be provided to registries.

4.2 Persistence

4.2.1 Principle

The principle of persistence ensures that the unique identifier remains associated with the same object at least throughout its life cycle. The purpose of a persistent identifier is to accurately represent an object for as long as it exists, regardless of changes in company status, ownership, location, including mergers, acquisitions, splits or spin-offs. The economic operator is responsible for recording, maintaining, and securely storing each identifier, either directly or through a DPP service provider, to ensure persistence.

4.2.2 Requirements

The unique identifier shall provide persistence, with responsibility assigned for:

1) *Consistency*: The unique identifier, once assigned, shall remain unchanged and consistently refer to the same object without ambiguity, throughout the life cycle of the object.

2) *Preservation:* The unique identifier shall be preserved, at least throughout its expected duration, or as long as required by applicable regulations.

3) *Permanence:* The unique identifier shall remain available, including in the event of insolvency, liquidation, or a cessation.

Note: For products placed on the European market, the EU DPP registry ensures a minimum level of preservation and permanence of identifiers.

4.3 Syntax

4.3.1 Principle

The principle of syntax establishes a clear, standardized structure for unique identifiers, ensuring consistent recognition and usage across systems, services, and platforms when needed. This structure supports interoperability and efficient processing, by enabling the mapping between different syntaxes. The unique identifiers maintain a distinguishable format to enhance identification, searchability and retrieval. The length is specified for optimized processing and storage, and the syntax structure enables accessibility and usability on smartphones, other similar consumer devices and common web browsers.

4.3.2 Requirements

The unique identifier shall follow a predefined syntax, by ensuring:

1) *Character set:* The unique identifier shall be a string of characters conforming to ISO/IEC 646:1991. If used as a URI, the string shall follow IETF, RFC 3986:2005 and keep unreserved characters as-is and percent-encode all other Unicode characters per ISO/IEC 10646:2020.

2) *Web use:* The unique product identifier shall either be provided in the form of a URL or derivable into a URL (e.g. resolvable URI to URL) through a specified conversion method according to the ID schemes specified in this document, thereby ensuring compatibility with web-based systems.

4.3.3 Guidelines

1) *Structure:* The unique identifier may conform to a structural pattern defined by the chosen ID scheme, supporting easier parsing and recognition across various systems, services and platforms.

2) *Variable length:* Unique identifiers may use variable length.

4.4 Granularity

4.4.1 Principle

The principle of granularity ensures that a unique identifier accurately represents the object it refers to, when needed. Identifiers are aligned with the characteristics of the objects they represent, including distinctions between different types of identifiers and varying levels of granularity. Additionally, identifiers facilitate access to a digital product passport, either directly, via a data carrier, or through a resolution mechanism.

4.4.2 Requirements

The unique product identifier shall maintain its intended meaning, by ensuring:

1) *Granularity:* Each product identifier shall be unique at least at the smallest granularity level it serves, whether it is a model, batch, or item.

2) *Granularity consistency:* The granularity level of a unique product identifier shall remain consistent once the product is placed on the market or put into service.

3) *Change in granularity:* If a change in granularity becomes necessary that cannot be accommodated with the existing identifier, a new or extended unique product identifier shall be assigned. The new or extended identifier shall be linked to the old unique identifier to maintain traceability.

4.4.3 Guidelines

1) *Granularity level:* In cases where regulatory requirements offer the choice for identification at the model, batch or item level, a more granular level of specificity for the unique product identifier (e.g. using batch instead of model, or item instead of batch) may be adopted to enhance traceability or meet specific operational needs.

2) *Resolver:* A resolver may be used for the resolution of a unique identifier.

3) *Hierarchy:* A unique product identifier may be structured to represent multiple levels of granularity, such as model, batch and individual item identification, within a single hierarchical format.

4.5 Interoperability

4.5.1 Principle

The principle of interoperability is enabled by the global uniqueness and the digital product system. In addition, unique identifiers ensure seamless operation across systems, services, and platforms without modification of the identifier or the platform, enabling consistent interpretation and unambiguous data exchange. A structured format supports compatibility with external components to enable automated processing and integration across industries.

Portability ensures continuity across technological and organizational transitions. Key interoperability interfaces include data carriers, web links, APIs, and digital product passports.

4.5.2 Requirements

The unique identifier shall be interoperable, by ensuring:

1) *Data carrier:* The unique product identifier shall be retrievable from a data carrier specified in EN 18220, linking it to a digital product passport.

2) *Digital access:* The unique product identifier shall be provided to retailers and online marketplaces for consumers to obtain the product's DPP.

3) *Compatibility:* The unique product identifier shall be encoded via a syntax that is compatible with external devices, such as professional scanning devices, smartphones and similar digital consumer devices.

4) *Existing ID:* Whenever possible, the use of existing ID schemes within one of the identification schemes specified in this document shall be enabled.

4.5.3 Guidelines

1) *Smartphones and similar devices:* When using state-of-the-art smartphones and similar consumer devices, the unique product identifier should be automatically useable and compatible with all major operating systems and browsers.

2) *Portability:* The unique identifier scheme should support intended systems, services, platforms or third-party without losing their meaning or function.

4.6 Openness

4.6.1 Principle

The principle of openness promotes transparency by ensuring that policies, processes, and standards are publicly accessible, allowing stakeholders to study and implement them. The use of widely accessible rules and procedures foster broad adoption and interoperability. Openness also facilitates the transition or switching between systems, services, service providers and platforms without compromising functionality or data integrity. Identifiers are used to access DPP public information without requiring registration, download DPP specific software, or authentication. Openness aims to avoid unnecessary restrictions that can possibly limit access, use, or innovation.

4.6.2 Requirements

The unique identifier shall remain open and transparent, by ensuring:

1) *Transparency:* The ID scheme and unique identifier shall be based on a system where policies, processes, and standards can be studied, influenced, and used under fair, reasonable, and non-discriminatory terms.

2) *Consumer usage:* To access public information, especially for consumer usage, the unique identifier shall be useable without the need to register, download DPP-specific software, or add user credentials.

3) *No undue restrictions:* The unique identifier shall not result in vendor lock-in. It shall also not impose limitations on use, except where such restrictions are technically necessary to ensure global uniqueness and interoperability.

5 ID schemes for products

5.1 General

A unique product identifier shall fulfil the requirements in Clause 4 and comply with one of the schemes in Clause 5. A product shall have a unique product identifier which shall be based on one of the ID schemes defined in Clause 5.

5.2 ID scheme 1: Web enabled, structured path and query ID for products

5.2.1 Description

This ID scheme ensures globally unique identification based on ISO/IEC 15459-4 (individual products) and ISO/IEC 15459-6 (grouped products). It structures identifiers using ISO/IEC 15418 and encodes them in URLs or resolves them via HTTP as per ISO/IEC 18975. Users can choose between two web-enabled approaches: a structured path format or a query string format. The scheme enables direct web access to digital product passports and supports offline parsing for supply chain and point-of-sale processes. This ID scheme is based on issuing agencies as per ISO/IEC 15459-2, ISO/IEC 15459-3 and ISO/IEC 15418.

5.2.2 Requirements

5.2.2.1 Web enabled, structured path ID for products

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to ISO/IEC 15459-4:2014 for individual products or ISO/IEC 15459-6:2014 for product groupings,

b) apply the GS1 Application Identifier (AI) or the ASC MH10.8.2 Data Identifier (DI) in accordance with ISO/IEC 15418:2016,

c) follow the structured path approach specified in ISO/IEC 18975:2024 for encoding and resolving identifiers over HTTPS,

d) conform with GS1, Digital Link Standard: URI Syntax, Version 1.6.0:2022 for users of GS1 Application Identifiers (AI), or

e) conform with EN IEC 61406-2:2024 or ISO/IEC 18975:2024 for users of ASC MH10.8.2 Data Identifiers (DI),

f) apply ISO/IEC 15459-3:2014 for global uniqueness, and

g) use a registered Issuing Agency Code according to ISO/IEC 15459-2:2015.

5.2.2.2 Web enabled, query string ID for products

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to ISO/IEC 15459-4:2014 for individual products and/or ISO/IEC 15459-6:2014 for product groupings,

b) apply the ASC MH10.8.2 Data Identifiers (DI) in accordance with ISO/IEC 15418:2016,

c) follow the query approach specified in ISO/IEC 18975:2024 for encoding and resolving identifiers over HTTPS, and

d) apply ISO/IEC 15459-3:2014 for global uniqueness, and

e) use a registered Issuing Agency Code according to ISO/IEC 15459-2:2015.

5.3 ID scheme 2: Identification Link (IL) string for products

5.3.1 Description

This ID scheme using Identification Links (IL) in URL format, based on EN IEC 61406-1, enables unique product identification on item level. Combining EN IEC 61406-1 and EN IEC 61406-2 extends the possible usage to product level, lot/batch level and item level, and allows usage to identify other entities such as organizations and locations. It allows economic operators to encode existing product ID schemes (model, batch, or item) directly into an IL string. IL strings are based on a self-issuing system, without external dependencies while supporting additional structuring via ASC MH10.8.2 Data Identifiers (DI) specified in ISO/IEC 15418. This approach ensures compatibility with existing ID schemes and enables direct web access to digital product passports.

5.3.2 Requirements

5.3.2.1 Item level identification (only)

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to EN IEC 61406-1:2022, with optional use of EN IEC 61406-2:2024, when a unique string of characters is sufficient, and no syntactical or semantic intelligence is required.

5.3.2.2 Model, batch and item level identification

To ensure compliance with this ID scheme, the following requirements shall be met:

a) use EN IEC 61406-1:2022 combined with EN IEC 61406-2:2024 when encoding standardized syntax and semantics is required.

b) include ASC MH10.8.2 Data Identifiers (DI) as specified in ISO/IEC 15418:2016 to define model, batch, and item identification.

5.3.3 Guidelines

The following should be used:

a) Identification Link according to EN IEC 61406-1:2022, where the string of characters of the IL should be short, e.g. to be encoded into a 96-bit RFID chip or to allow a compact 2D-code for small products; or where protection against the threat of counting up the product serial number shall be intercepted by masking model and serial number with a random alphanumerical string.

b) Identification Link according to EN IEC 61406-2:2024 for comprehensive product identification across model, batch, and item levels, ensuring semantic interoperability with existing ID schemes. It also can contain internal further information about the kind of identified object, for example product, person, location or document or additional data elements or classifications.

5.4 ID scheme 3: Decentralized identifiers (DID) for products

5.4.1 Description

This ID scheme is a self-issuing system that utilizes Decentralized Identifiers (DIDs) for self-generated, globally unique identification. DIDs support cryptographic authentication and privacy

preserving information management. The scheme ensures interoperability with centralized, federated, and decentralized ID systems. Methods such as did:web, did:ethr, and did:ebsi provide flexible implementations, with did:web offering a lightweight, non-DLT solution where did:ebsi/did:ethr enable dynamic updates of deep links to digital product passports, e.g. in the case the economic operator responsible for the DPP changes. A DID can represent any level of granularity (batch, model, item). This approach enhances security, flexibility, and adaptability for digital identity management.

5.4.2 Requirements

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to W3C, Decentralized Identifiers (DIDs) v1.0:2022 to ensure proper DID syntax, core properties, and resolving DIDs to the resources it represent.

5.4.3 Guidelines

One of the following DID methods should be used:

a) DID [7],

b) DID [8] or

c) DID [9].

5.5 ID scheme 4: Identification for products and product groups

5.5.1 Description

This ID scheme is suitable for limited capacity data carriers, namely RAIN RFID (UHF RFID) per ISO/IEC 18000-63 and 2D codes, that utilize resolvers to access DPP. The method ensures globally unique identification using ISO/IEC 15459-4 (individual products) and ISO/IEC 15459-6 (grouped products). This scheme structures and encodes data for RFID applications based on the ISO/IEC 15961 series, ISO/IEC 15962, ISO/IEC 17360 and EPC TDS 2.3, and uses ISO/IEC 15434 for encoding data in 2D codes. The scheme standardizes application communication with data processors and operates independently of the RAIN RFID (UHF RFID) per ISO/IEC 18000-63. It supports offline processing of product details and follows ISO/IEC 15459-2, ISO/IEC 15459-3 and ISO/IEC 15418 for identifier allocation and syntax. For additional security, the economic operator may optionally apply ISO/IEC 20248 [10] for digital signatures. URL-strings can optionally be embedded as data elements.

5.5.2 Requirements

5.5.2.1 Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000-63

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to ISO/IEC 15459-4:2014 for individual products, and/or ISO/IEC 15459-6:2014 for groupings of products, to ensure the uniqueness of product identifiers,

b) follow ISO/IEC 15961-1:2021 or ISO/IEC 17360:2023 or GS1, EPC Tag Data Standard (TDS), Version 2.3:2025 for semantic structuring and logical representation of data, and

c) apply ISO/IEC 15459-3:2014 for global uniqueness, and

d) use a registered Issuing Agency Code according to ISO/IEC 15459-2:2015.

5.5.2.2 Identification for products and product groups, 2D-symbols

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to ISO/IEC 15459-4:2014 for individual products, and/or ISO/IEC 15459-6:2014 for groupings of products, to ensure the uniqueness of product identifiers,

b) use the ASC MH 10.8.2 Data Identifier according to ISO/IEC 15418:2016 for the semantics,

c) structure the data transfer syntax for high-capacity automatic data capture media according to ISO/IEC 15434:2025 or DIN 16598:2022 when encoding structured data identifiers,

d) apply ISO/IEC 15459-3:2014 for global uniqueness, and

e) use a registered Issuing Agency Code according to ISO/IEC 15459-2:2015.

5.5.3 Guidelines

One or more of the following may additionally be used:

a) ISO/IEC 15962 [11] for RAIN RFID (UHF RFID) per ISO/IEC 18000-63 data encoding and decoding,

b) ISO/IEC 18975:2024 for the format of HTTP URI to be encoded in RAIN RFID (UHF RFID) per ISO 18000-63, and

c) ISO/IEC 20248 [10] for digital signatures.

5.6 ID scheme 5: Digital Object Identifier (DOI) for products

5.6.1 Description

This ID scheme, based on ISO 26324, enables unique identification of products using Digital Object Identifiers (DOI) in URL format, consistent with the structured path approach. It allows economic operators to encode existing product IDs (model, batch, or item) directly into the DOI. The DOI is based on a flexible system of either self-issuing or assignment by a service provider depending on the practices of the enabling service provider. This ID scheme is based on issuing agencies as per ISO 26324 and further documented in the DOI Handbook [12]. It allows compatibility with existing ID schemes and enables direct web access to digital product passports.

5.6.2 Requirements

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to ISO 26324:2025 for model, batch or item identification,

b) declare whether the DOI applies to the product model, batch or item, and

c) use lowercase doi.org at the beginning of a DOI.

6 ID schemes for economic operators and facilities

6.1 General

A unique economic operator identifier and unique facility identifier shall fulfil the requirements in Clause 4 and comply with one of the schemes in Clause 6, when they are required by regulation.

An economic operator or facility shall have a unique economic operator identifier or facility identifier which shall be based on one of the ID schemes defined in Clause 6.

6.2 ID scheme 6: Economic operator identification using ISO/IEC 6523

6.2.1 Description

This ID scheme is based on ISO/IEC 6523-1, providing a standardized framework for unique identification of economic operators. The scheme allows hierarchical assignment of identifiers to economic operators, enabling companies to allocate unique IDs to parent entities, subsidiaries, operational divisions, factories, warehouses, and offices who are acting in the value chain. ISO/IEC 6523 ICD Prefixes provide data system uniqueness between identification schemes whose structure, format, and syntax is defined by the organization for each registered scheme.

6.2.2 Requirements

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to ISO/IEC 6523-1:2023 to ensure uniqueness between economic operator identifier schemes,

b) use a registered ID scheme according to ISO/IEC 6523-2:2025,

c) for constructing and resolving economic operator and facility identifiers, conform to the specifications of the ISO/IEC 6523 registered ID scheme.

6.2.3 Guidelines

Examples of ISO/IEC 6523 registered ID schemes (see C.3.1 for more examples):

— ISO 17442-1 [13] for Legal Entity Identifiers (LEI) for economic operators,

— ASC MH10.8.2 Data Identifier (DI) qualified identifiers for economic operators,

— GS1 Application Identifier (AI) qualified identifiers for economic operators,

— PEPPOL-ID, from the Pan-European Public Procurement Online network for economic operators.

If instrumental, the economic operator may additionally use:

a) ISO/IEC 18975:2024 for encoding and resolving identifiers over HTTP, or

b) EN IEC 61406-2:2024 for constructing identifiers as URLs.

6.3 ID scheme 7: Economic operator and facility identification per ISO/IEC 15418

6.3.1 Description

This ID scheme, originally created for unique identification used within automatic identification and data capture technology in the value chain, is based on ISO/IEC 15418 and ISO/IEC 15459 or alternatively for MH10.8.2 Data Identifier (DI) users EN IEC 61406-2 System Identifier and Data Identifier (SIDI), for encoding unique and interoperable identifiers within DPP data carriers. The scheme allows hierarchical assignment of identifiers, enabling companies to allocate unique IDs to actors in the value chain such as parent entities, subsidiaries, and operational divisions, factories, warehouses, and offices and to physical locations such as a factory where production of a product occurred. It facilitates access to organizational data via data carriers and supports offline parsing of identifiers for system integration and backup processes. ISO/IEC 15459 Issuing

Agency Codes (IACs) provide uniqueness data system between identification schemes whose structure, format, and syntax is defined by the organization for each registered scheme.

ISO/IEC 15459-2 Issuing Agency Codes or alternatively for MH10.8.2 Data Identifier (DI) users EN IEC 61406-2 provides uniqueness between identification schemes used in data carriers and/or data systems whose structure, format, and syntax is defined by the organization for each registered scheme, promoting global interoperability in supply chain and administrative workflows.

6.3.2 Requirements

To ensure compliance with this ID scheme, the following requirements shall be met:

a) apply the GS1 Application Identifier (AI) or the ASC MH10.8.2 Data Identifier (DI) in accordance with ISO/IEC 15418:2016,

b) for ASC MH10.8.2 Data Identifier (DI), conform to EN IEC 61406-2:2024, or

c) for ASC MH10.8.2 Data Identifier (DI), conform to ISO/IEC 15459-3:2014, ISO/IEC 18975:2024, and use a registered ID scheme according to ISO/IEC 15459-2:2015 for encoding economic operators and facility identifiers, or

d) for GS1 Application Identifier (AI), conform to GS1, General Specifications Standard, Version 25.0:2025 per ISO/IEC 15459-3:2014, GS1, Digital Link Standard: URI Syntax, Version 1.6.0:2022 per ISO/IEC 18975:2024, and use a GS1 registered ID scheme according to ISO/IEC 15459-2:2015 for encoding economic operators and facility identifiers.

6.4 ID scheme 8: Economic operator identification in data systems using Decentralized identifiers (DID)

6.4.1 Description

This Decentralized Identity (DID) framework is a self-issuing system that provides a globally unique, self-sovereign identifier that is based on W3C Decentralized Identifiers (DIDs). Each DID resolves to a public DID document that defines its properties and contains a public key for verification. A DID becomes an Economic Operator Identifier when an authorized company register issues a Verifiable Credential (VC) to it. This enables economic operators to sign and issue verifiable, tamper-proof digital product passports. The public key for verification can be found in the public DID document, ensuring transparency and trust. Decentralized Authentication and Access Control can be implemented via Role Verifiable Credentials issued to the economic operator DID.

6.4.2 Requirements

To ensure compliance with this ID scheme, the following requirements shall be met:

a) adhere to W3C, Decentralized Identifiers (DIDs) v1.0:2022 for creation of self-sovereign identities including public-private key pairs,

b) follow W3C, Verifiable Credentials Data Model v2:2025 to associate verifiable credentials to the economic operator DID or the facility DID, and

c) use W3C, Decentralized Identifier Resolution (DID Resolution) v0.3:2025 to find the DID document.

6.4.3 Guidelines

If instrumental, the following may additionally be used:

a) use trusted company registries, for example, national company registries or global company registries, such as the Global Legal Entity Identifier Foundation (GLEIF). To bind a DID to a company, GLEIF would issue a vLEI (Verifiable Legal Entity Identifier) [14] based on a national company register to the DID at hand.

b) associate other relevant Verifiable Credentials with the economic operator DID—such as legal person identifier (LPID), which is a verifiable credential within the European Digital Identity Wallet (EUDI Wallet), sustainability disclosures, compliance reports, or ISO certifications—to create a chained provenance trail.

6.5 ID scheme 9: Economic operator identification used in data systems per Digital Object Identifiers (DOI)

6.5.1 Description

This ID scheme, based on ISO 26324, enables unique identification of organizations and facilities using the Digital Object Identifiers (DOI) in URL format. This ID scheme is based on issuing agencies as per ISO 26324 and further documented in the DOI Handbook. It allows the use of existing identifiers or allocation of new identifiers depending on the practices of the issuing agency. It allows compatibility with existing ID schemes and enables direct web access to information about the identified object.

6.5.2 Requirements

To ensure compliance with this ID scheme, the following requirements shall be met by the registrant:

a) adhere to ISO 26324:2025 to ensure unique economic operator and facility identifiers where sufficient metadata are required by the issuing agency to ensure uniqueness of the organization or facility to which the identifier is assigned, and

b) use DOI resolution to access the digital product passport.

Annex A
(informative)

Selecting identification granularity levels for products

A.1 Scope

This annex provides guidance to assist economic operators in determining the appropriate level of product identification granularity — model, batch, or item — prior to market release. In cases where regulatory requirements allow identification at different granularity levels, the decision should balance regulatory, technical, ecological, and economic considerations.

A.2 Purpose of this guidance

This guidance aims to streamline the decision-making process regarding granularity levels of product identification, ensuring compliance with regulatory requirements while optimizing ecological and economical outcomes.

A.3 General principles

Selecting the appropriate level of identification can lead to significant efficiency gains and reductions in cost and effort throughout the product's life cycle. If DPPs are required only at the model level, identification on batch-level or item-level can still be appropriate when future use can be expected. Life cycle stages, such as downstream handling, can also determine the need for batch or item-level identification.

A.4 Granularity levels

Identification granularity can be as follows:

a) *Model-level identification*; Model-level identification is generally suitable for most products. It supports inventory management, product classification, and alignment of product information between economic operators.

b) *Batch-level identification*; Batch-level identification can be appropriate when it is necessary to trace a set of products produced in a manufacturing location with similar characteristics but no further distinction between individual items is required.

c) *Item-level identification*; Item-level identification can be applied when detailed information on individual products is required, or when justified by specific product characteristics, use cases, or business needs.

A.5 Conditions for considering item-level identification

Identification at the item level is advisable for products that meet any of the following criteria:

— when item-level identification can be expected to be used after the product is placed on the market,

— designed to be repaired, refurbished, or overhauled by changing parts or components,

— capable of being upgraded or enhanced by changing parts or components,

— equipped with embedded software that can be updated,

— subject to regular maintenance, inspection, or statutory audit,

— where wear level, operating hours, or other usage information needs to be recorded,

— requiring unique identification on an individual basis to fulfil the intended purpose,

— exposed to a high risk of counterfeiting, and

— if gathering data on the installed product base for strategic product planning purposes is needed.

A.6 Change in granularity

In cases where finer granularity becomes necessary and was not foreseen, a product identified at the model level can later be identified at the batch or item level. When this occurs, a new or extended unique product identifier can be assigned, where the new identifier is linked to the previous one to maintain traceability and preserve the connection to the original product information.

Annex B
(informative)

Overview of ID schemes for products

B.1 General information for ID schemes

B.1.1 Governance and uniqueness

See Table B.1 for general information about governance and uniqueness on ID schemes for products.

Table B.1 — Governance and uniqueness on ID schemes for products

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
Identifier standard	ISO/IEC 15459 + ISO/IEC 15418 (GS1 AI or ASC MH10.8 DI)	ISO/IEC 15459 + ISO/IEC 15418 (ASC MH10.8 DI)	EN IEC 61406-1	EN IEC 61406-1 + EN IEC 61406-2	DIDs v1.0:2022.	ISO/IEC 15459 + ISO/IEC 15418 (GS1 AI or ASC MH10.8 DI)	ISO/IEC 15459 + ISO/IEC 15418 (ASC MH10.8 DI)	ISO 26324
Type of issuing	Issuing agency ISO/IEC 15459 issuing agency then self-issuing by economic operator	Issuing agency ISO/IEC 15459 issuing agency then self-issuing by economic operator	Self-issuing system Domain registration authority then self-issuing by economic operator	Self-issuing system Domain registration authority then self-issuing by economic operator	Self-issuing system ICANN prefix then self-issuing by economic operator	Issuing agency ISO/IEC 15459 issuing agency then self-issuing by economic operator	Issuing agency ISO/IEC 15459 issuing agency then self-issuing by economic operator	Issuing agency ISO/IEC 15459 issuing agency code then self issuing by economic operator

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Uniqueness	ISO/IEC 15459		Web domain (RFC 3986)	Web domain (RFC 3986), optionally with ISO/IEC 15459 or other ID	DIDs v1.0:2022	ISO/IEC 15459		ISO 26324 + ISO/IEC 15459.
Reliance of domain name (for the unique identifier)	No, unique identifier is independent of domain name.	No, unique identifier is independent of domain name.	Yes, uses domain name as part of the identifier for global uniqueness.	Yes, uses domain name as part of the identifier for global uniqueness.	No, identifier is self-resolving using DID method syntax.	No, unique identifier is independent of domain name.	No, unique identifier is independent of domain name.	Yes, relies on centralized DOI resolution via domain infrastructure.

B.1.2 Syntax and semantics

See for Table B.2 general information about syntax and semantics on ID schemes for products.

Table B.2 — Syntax and semantics on ID schemes for products

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
Technical standard for syntax	ISO/IEC 18975 + ISO/IEC 15459-2 and −3 + RFC 3986		RFC 3986			RFC 3986 (ISO/IEC 1800 0-63)	RFC 3986	ISO 26324 + ISO/IEC 15459 -2 and -3

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Application standard syntax	ISO/IEC 18975 or EN IEC 61406-2 for ASC MH.10.8.2 DIs. ISO/IEC 18975 and GS1 Digital Link URI for GS1 AIs	AutoID-URL 1.3 of ISO/IEC 18975 (compatible with query string of ISO/IEC 18975)	EN IEC 61406-1	EN IEC 61406-1 and EN IEC 61406-2 (compatible with query string of ISO/IEC 18975)	DIDs v1.0:2022	ISO/IEC 15961 series, ISO/IEC 15962, ISO/IEC 17360, ISO/IEC 20248, or EPC TDS 2.3.	ISO/IEC 20248, DIN 16598 (ASC MH10.8.2 DI)	ISO 26324
Technical standard for semantics	ISO/IEC 15459 + ISO/IEC 18975 + ISO/IEC 15418 (AI or DI)	ISO/IEC 15459 + ISO/IEC 15418 (DI) + ISO/IEC 18975	N/A, structure-only, no embedded semantics	ISO/IEC 15418 (ASC MH10.8.2 DI)	DIDs v1.0:2022	ISO/IEC 15459, ISO/IEC 15962, ISO/IEC 18975.	ISO/IEC 15459, ISO/IEC 15418 (ASC MH10.8.2 DI), DIN 16598.	ISO 26324
Application standard for semantics	ISO/IEC 15418 (AI or DI)	ISO/IEC 15418 (DI)	N/A – no embedded semantics	EN IEC 61406-2, DI of ISO/IEC 15418, optionally ISO/IEC 15459 DI qualified identifier can be used.	DIDs v1.0:2022	ISO/IEC 15961 series, ISO/IEC 17460, ISO/IEC 18975, ISO/IEC 20248	ISO/IEC 15418, DIN 1659	ISO 26324

B.1.3 Security

See for Table B.3 general information about security on ID schemes for products.

Table B.3 — Security on ID schemes for products

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
Can be combined with ISO/IEC 20248	Yes, supports digital signature for data authentication.				Yes, for when the DID is encoded in RAIN RFID (UHF RFID) per ISO/IEC 18000 -63, or when a separate data carrier is used from a DID barcode.	Yes, supports digital signature for data authentication (tags with additional data capacity).	Yes, supports digital signature for data authentication.	Yes, for when the DOI is encoded in RAIN RFID (UHF RFID) per ISO/IEC 18000 -63, or when a separate data carrier is used from a DOI barcode.
Can be combined with ISO 22376 [15]	Yes, for ASC MH10.8.2 (DI structure + tagged data). No for GS1 AIs.	Yes, compatible for data element structure and transfer syntax.			No	Yes, for ASC MH10.8.2 (DI structure + tagged data). No for GS1 AIs.	Yes, compatible for data element structure and transfer syntax.	

B.1.4 Issuing process and prerequisites

See Table B.4 for general information about issuing process and prerequisites of ID schemes for products.

Table B.4 — Issuing process and prerequisites on ID schemes for products

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link	Identification link (structured ID link)	Decentralized identifer	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier
Prerequisite for web enabled access to DPP	Registered web domain encoded or discovered via use of resolution service				Registered web domain	Registered web domain encoded or discovered via use of resolution service		
Process of issuing of product identifier	1. Get CIN (ISO/IEC 15459 issuing agency) 2. Issue product ID within assigned number band.		1.) Issue ID based on web domain.		For did:web: 1.) Use web domain 2.) Create DID 3.) Publish DID document	1. Get CIN (ISO/IEC 15459 issuing agency) 2. Issue product ID within assigned number band		1. Get CIN (ISO/IEC 15459 issuing agency) 2. Issue product ID within assigned number band either by operator or issuing agency

ID scheme	ID scheme 1	ID scheme 2	ID scheme 3	ID scheme 4	ID scheme 5
Sovereignty about the data in the product identifier	Economic operator controls ID elements	Economic operator controls self-issued elements	Full control; verifiable via cryptography	Economic operator controls product IDs	
Changes in technology that would affect the product identifier	Changes in ISO/IEC standards or RFC 3986				
Procedures for switching a service provider / issuing agency	No restrictions	No restrictions. Domain transfer is automated.	No restrictions		No restrictions. Switching between agencies allowed.

B.2 Requirements and guidelines

B.2.1 Global uniqueness

See Table B.5 for information on requirements and guidelines of 4.1.

Table B.5 — Information on requirements and guidelines for global uniquness

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
4.1.2 (1) No reassignment	Yes, ensured by the rules set in ISO/IEC 15459.		Yes, ensured by the rules in EN IEC 61406-1.	Yes, ensured by the rules in EN IEC 61406-1 and EN IEC 61406-2.	Yes, ensured by W3C Decentralized Identifiers (DIDs) v1.0	Yes, ensured by the rules set in ISO/IEC 15459.		Yes, each ID is registered with unique metadata (checked by issuing agency).
4.1.2 (2) Distinct	Yes, globally unique and context independent per ISO/IEC 15459-3.		Yes, the web domain remains the anchor for global uniqueness.		Yes, globally unique per W3C DIDs v1.0:2022.	Yes, globally unique per ISO/IEC 15459-3.		Yes, globally unique per ISO/IEC 15459 -3 and ISO 26324.

ID scheme	ID scheme 1	ID scheme 2	ID scheme 3	ID scheme 4	ID scheme 5
4.1.2 (3) Non-coexistence	Yes, prohibits reusing or reassigning identifiers.		Yes, prohibits reusing or reassigning identifiers ensured by cryptography and DID method rules.	Yes, prohibits reusing or reassigning identifiers.	
4.1.2 (4) Cross domain identification	5.2	5.3	5.4	5.5	5.6

B.2.2 Persistence

See Table B.6 for information on requirements and guidelines of 4.2.

Table B.6 — Information on requirements and guidelines for persistence

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products

ID scheme	ID scheme 1	ID scheme 2		ID scheme 3	ID scheme 4	ID scheme 5
4.2.2 (1) Consistency	Yes, ISO/IEC 15459-3.	Yes, ensured by the rules in EN IEC 61406-1 and the EU DPP System Registry.	Yes, ensured by the rules in EN IEC 61406-1 and EN IEC 61406-2 and the EU DPP system registry.	Yes, ensured by W3C DID v1.0 – persistence is technology-driven, not process-based.	Yes, ISO/IEC 15459-3.	Yes, ensured by requiring descriptive metadata for each entity provided to issuing agency.
4.2.2 (2) Preservation	Yes, supported by life cycle management rules per ISO/IEC 15459-3.	Yes, ensured by life cycle management rules per EN IEC 61406-1 and the EU DPP System.	Yes, ensured by the life cycle management rules per EN IEC 61406-1 and −2 and the EU DPP System.	Yes, through decentralized structure and governance throughout the life cycle.	Yes, supported by life cycle management rules per ISO/IEC 15459-3.	Yes, issuing agency responsible for preservation of identifiers.
4.2.2 (3) Permanence	Yes, per ISO/IEC 15459-3 and issuing agency rules.	Yes, ensured per EN IEC 61406-1 and −2 and by the EU DPP system. Web domains can be transferred or redirected.		Yes, under the control of the DID owner.	Yes, per ISO/IEC 15459-3 and issuing agency rules.	Yes, issuing agencies require the identifiers remain associated with the described referent.

B.2.3 Syntax

See Table B.7 for information on requirements and guidelines of 4.3.

Table B.7 — Information on requirements and guidelines for syntax

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
4.3.2 (1) Character set	ISO/IEC 646, IETF RFC 3986 and ISO/IEC 10646.						ISO/IEC 15434 or DIN 16598.	ISO 26324.
4.3.2 (2) Web use	IETF RFC 3986 (URI HTTP and URL) + additional rules set in ISO/IEC 18975		IETF RFC 3986 (URL) + additional rules set in EN IEC 61406-1	IETF RFC 3986 (URL) + additional rules set in EN IEC 61406-1 and EN IEC 61406-2	IETF RFC 3986 (URI)	Not applicable (not web-enabled)		IETF RFC 3986 (URL)
4.3.3 (1) Structure	ISO/IEC 15459 and ISO/IEC 15418 AI or DI	ISO/IEC 15459 and ISO/IEC 15418 DI	Can optionally integrate ISO/IEC 15459 (DIs), GUIDs, DIDs, or internal IDs (e.g. product name + serial number)		"did": prefix with method (e.g. web, ethr, ebsi) and method-specific ID	ISO/IEC 15459 ISO/IEC 15961 series ISO/IEC 15962 ISO/IEC 17360, ISO/IEC 18000 -63 and EPC TDS v.2.3	ISO/IEC 15434 (ASC MH10.8.2 DI only)	ISO 26324 – issued prefix + constructed suffix (by operator or agency)

4.3.3 (2) Variable length	Yes, combination of fixed, structured, and variable elements.	Yes	Yes, variable between 6 and 255 characters.	Yes	Yes, combination of fixed, structured, and variable elements.	Yes, can be fixed depending on issuing agency DOI foundation rules.

B.2.4 Granularity

See Table B.8 for information on requirements and guidelines of 4.4.

Table B.8 — Information on requirements and guidelines for granularity

ID scheme	**ID scheme 1**		**ID scheme 2**		**ID scheme 3**	**ID scheme 4**		**ID scheme 5**
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
4.4.2 (1) Granularity	Model, batch, item via qualifier	Model, batch, item via ASC MH 10.8.2 DI	Item level only. Granularity also represented in graphical marking of data carrier.	Model, batch, item via Data Identifier. Granluarity also represented in graphical marking of data carrier.	Model, batch, item as DID document attributes	Model, batch, item via qualifier		Model, batch, item

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4	ID scheme 5
4.4.2 (2) Granularity consistency	Supports multiple levels.		Item level only	Supports multiple levels.	Granularity set in DID document.	Supports multiple levels.	Supports multiple levels, fixed once registered.
4.4.2 (3) Change in granularity	Change in granularity requires new or extended identifier.		No need for granularity change as already item-level.	Change in granularity requires new or extended identifier.	Allowed – managed via DID document.	Change in granularity requires new or extended identifier.	Change in granularity requires new identifier.
4.4.3 (1) Granularity level	Allowed	Flexible	Only item level	Allowed	All levels supported.	Allowed	Flexible
4.4.3 (2) Resolver	Browser-based, no special resolver				DID resolver + DNS (per W3C spec).	Browser-based, no special resolver	HTTP resolution per ISO 26324
4.4.3 (3) Hierarchy	Yes		No	Yes			Per issuing agency

B.2.5 Interoperability

See Table B.9 for information on requirements and guidelines of 4.5.

Table B.9 — Information on requirements and guidelines for interoperability

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
4.5.2 (1) Data carrier	Web-enabled data carriers					RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	2D-symbols	Web-enabled data carrier
4.5.2 (2) Digital access	Yes							
4.5.2 (3) Compatibility	Yes, can be used with web technologies and existing supply chain equipment as per ISO/IEC 15459.		Yes, can be used with web technologies and existing supply chain equipment.			Yes, can be used with web technologies and existing supply chain equipment as per ISO/IEC 15459.		
4.5.2 (4) Existing ID	Yes							

ID scheme	ID scheme 1	ID scheme 2	ID scheme 3	ID scheme 4		ID scheme 5
4.5.3 (1) Smartphones and similar devices	2D/NFC via URLs, smartphone compatible and readable in browsers		DID links via 2D/NFC, smartphone compatible and readable in browsers	Using second data carrier	Using second data carrier	2D/NFC via URLs, browser-compatible
4.5.3 (2) Portability	Yes, globally unique, cross-platform.					

B.2.6 Openness

See Table B.10 for information on requirements and guidelines of 4.6.

Table B.10 — Information on requirements and guidelines for openness

ID scheme	ID scheme 1		ID scheme 2		ID scheme 3	ID scheme 4		ID scheme 5
Reference	**5.2.2.1**	**5.2.2.2**	**5.3.2.1**	**5.3.2.2**	**5.4.2**	**5.5.2.1**	**5.5.2.2**	**5.6.2**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link for products	Identification link for products (structured ID link)	Decentralized identifer for products	Identification for products and product groups, RAIN RFID (UHF RFID) per ISO/IEC 18000 -63	Identification for products and product groups, 2D-symbols	Digital object identifier for products
4.6.2 (1) Transparency	Public ISO/IEC and web standards		Public EN IEC and web standards		Public W3C standards	Public ISO/IEC standards	Public ISO/IEC and DIN standards	Public ISO + ISO/IEC standards
4.6.2 (2) Consumer usage	No additional software needed.					IDs may need to be parsed and resolved.		No additional software needed.

ID scheme	ID scheme 1	ID scheme 2	ID scheme 3	ID scheme 4	ID scheme 5
4.6.2 (3) No undue restrictions	No undue restrictions.	No undue restrictions, only requires domain registration, which is transferable between providers.	No undue restrictions, DIDs are fully controlled by the owner and do not rely on any third party.	No undue restrictions.	No undue restrictions, operator can switch issuing agencies or methods; existing IDs remain valid.

B.3 Examples

B.3.1 ID scheme 1: Web enabled, structured path and query string identification for products

B.3.1.1 Web enabled, structured path identification for products

See Table B.11 for an example of 5.2.2.1 using GS1 AI per ISO/IEC 15418.

Table B.11 — Example of web enabled structured path identification for products

Component	Meaning	Value
GTIN (01)	Global Trade Item Number	09524000059109
Consumer product variant (22)	Used to distinguish variants	2A
Batch number (10)	Batch identifier	ABC123
Serial Number (21)	Unique item-level identifier	12345XYZ
Production date (11)	Format YYMMDD	251121
Data carrier	2D code example	
URL scheme	https://example.com	
Full Example	https://example.com/01/09524000059109/22/2A/10/ABC123/21/12345XYZ?11=251121	
NOTE The above given example for a "Web enabled, structured path identification for products", is at the same also an "Identification Link" conform to EN IEC 61406-1.		

B.3.1.2 Web enabled, query string identification for products

See Table B.12 for an example of 5.2.2.2 using ASC MH10.8.2 DI per ISO/IEC 15418.

Table B.12 — Example of web enabled query string identification for products

Component	Meaning	Value
Product ID (.25P)	IAC + CIN + Product Number	QCELMI12345
Serial number (.S)	Serial number	654321
Data carrier	2D code example	
URL scheme	https://example.com	

Component	Meaning	Value
Full Example	https://www.example.com/?.25P=QCELMI12345&.S=654321	
NOTE The above given example for a "Web enabled, query string identification for products", is at the same also an "Identification Link" conform to EN IEC 61406-1, and EN IEC 61406-1 combined with EN IEC 61406-2.		

B.3.2 ID scheme 2: Identification Link (IL) string for products

B.3.2.1 Item level identification (only)

See Table B.13 for an example of 5.3.2.1 using EN IEC 61406-1.

Table B.13 — Example of Identification Link (IL) without structure

Component	Meaning	Value
Identifier (freetext)	Unstructured ID string (EN IEC 61406-1 approach)	freetext
Data carrier	2D code example with graphical frame to indicate item.	
URL scheme	https://www.example.com	
Full example	https://www.example.com/freetext	

B.3.2.2 Model, batch and item level identification

See Table B.14 for an example of 5.3.2.2 using EN IEC 61406-1, EN IEC 61406-2, and ASC MH10.8.2 DI per ISO/IEC 15418.

Table B.14 — Example of Identification Link with serial number per ISO/IEC 15418

Component	Meaning	Value
URL scheme	Protocol and domain (EN IEC 61406-2 approach)	https://www.example.com
Serial Number (.S)	Free-format or MH10.8.2 (DI) per ISO/IEC 15418	freetext
Batch Number (.10B)	Batch or lot number	freetext
Data carrier	2D code example with graphical frame to indicate batch or item.	
Full example:	https://www.example.com/?.S=freetext&.10B=freetext	

See Table B.15 for an example of 5.3.2.2 using EN IEC 61406-1 and EN IEC 61406-2.

Table B.15 — Example of Identification Link for model identification

Component	Meaning	Value
URL scheme	Protocol and domain (EN IEC 61406-2 approach)	https://www.domain-abc.com
Product Code (.P)	Model-level identifier	freetext *(or from ID scheme)*
Data carrier	2D code example with graphical frame to indicate model.	
Full example	https://www.example.com/?.P=freetext	
NOTE Product code level with optional reuse of existing ID solutions.		

B.3.3 ID scheme 3: Decentralized identifiers (DIDs) for products

See Table B.16 for an example of 5.4.2 Decentralized Identifiers (DIDs) v1.0 and did:web.

Table B.16 — Example of decentralized identifiers (DIDs) for products

Component	Meaning	Value
Product DID	Full DID identifying a product (DID method = web)	did:web:example.com:model4TR
Service type	Optional service query parameter specifies what to retrieve	?service = item-dpp
DID Method	Method part of the DID determines resolution rules	web
Resolution Standard	Resolver must follow Decentralized Identifier Resolution (DID Resolution) v0.3:2025	W3C-compliant
Returned Resource	Web resource	Digital product passport
Data carrier	2D code example	
URI scheme	https://resolver.io (DID resolver domain used to resolve the DID)	
Full example	https://resolver.io/did:web:example.com:model4TR/?service=item-dpp	

B.3.4 ID scheme 4: Product and group identification

B.3.4.1 Product and group identification, RAIN RFID (UHF RFID) per ISO/IEC 18000-63

See Table B.17 for example of 5.5.2.1 using ISO/IEC 15961-2 and ISO/IEC 17360.

Table B.17 — Example of identification based on ISO-registered ISO/IEC 15961-2

Component	Meaning	Value
RAIN MB01 (UII/EPC) encoding	Full encoded UII including AFI, DI, IAC, CIN, SN	AFI = 0xA1, “25SUN043325711MH80 31200000000001”
UII type (AFI = 0A1)	Indicates ISO/IEC 17360 6-bit character encoding	ISO/IEC 17360 6-bit encoded
UII purpose	Identifies a serialised product issued by an ISO/IEC 15459 registered entity	DI = 25S
Entity CIN issuing agent	ISO/IEC 15459 Issuing Agency Code (IAC)	UN (Dun and Bradstreet)
Entity CIN	Company Identification Number assigned by D&B	043325711
Serial number	Unique serial number per item	MH8031200000000001

See Table B.18 for example of 5.5.2.1 using GS1 SGTIN-96 EPC encoding.

Table B.18 — Example of identification based on ISO/IEC 15459-2 GS1 SGTIN-96 EPC encoding

Component	Meaning	Value
GTIN	Global Trade Item Number	09506000134352
Serial Number	Unique serial number for the item	123456789
EPC Pure Identity URI	Carrier-neutral URN representation for legacy data sharing	urn:epc:id:sgtin:9506000 1343.05.123456789
EPC Tag URI	RFID tag-specific URN representation	urn:epc:tag:sgtin-96:3.95060001343.05.12 3456789
EPC Hex	Binary encoding on RFID tag, beginning with address 20h of MB01	3066C4409047E140075B CD15

See Table B.19 for example of 5.5.2.1 using GS1 SGTIN-96 EPC (per GS1 TDS).

Table B.19 — Example of identification based on ISO/IEC 15459-2 GS1 SGTIN-96 EPC encoding (GS1 TDS)

Component	Meaning	Value
GTIN	Global Trade Item Number	09506000134352
Serial Number	Unique serial number for the item	123456789

Component	Meaning	Value
GS1 Digital Link URI	Carrier-neutral representation for data sharing via resolvable Web URI	https://id.gs1.org/01/09506000134352/21/123456789
EPC Hex	Binary encoding on RFID tag, beginning with address 20h of MB01	F7009506000134352091D6F3454

See Table B.20 for example of 5.5.2.1 using GS1 SGTIN++ EPC encoding

Table B.20 — Example of identification based on ISO/IEC 15459-2 for GS1 SGTIN++ EPC encoding

Component	Meaning	Value
GTIN	Global Trade Item Number	79521141123453
Serial Number	Unique serial number for the item	32a%2Fb
GS1 Digital Link URI	Carrier-neutral representation for data sharing via resolvable Web URI	https://example.com/01/79521141123453/21/32a%2Fb
EPC[SG1] Hex**	Binary encoding on RFID tag, beginning with address 20h of MB01	FD379521141123453856 6CB0AFC525065F1876F0D996D800

B.3.4.2 Product and group identification, 2D-symbols

See Table B.21 for an example of 5.5.2.2 using ACS MH10.8.2 DI per ISO/IEC 15434 (Format 6).

Table B.21 — Example of product and group identification, 2D-symbols

Component	Meaning	Value
Barcode encoding (ISO/IEC 15434)	Full encoded message using Format 06 (DI encoding). RS = Record Separator, GS = Group Separator	[) > RS06GS25SUN043325711MH8031200000000001GS1T110780GSQ21GS4LUSEOT
UII type	ISO/IEC 15434 Format 06, which indicates Data Identifier (DI)–based encoding	ISO/IEC 15434 Format 06 (DI encoding)
UII purpose	Identifies a serialized product issued by an ISO/IEC 15459-registered entity	DI = 25S
Entity CIN issuing agent	Issuing Agency Code (IAC) that assigns CINs	UN (Dun and Bradstreet)
Entity CIN	Company Identification Number assignable via D&B database lookup	043325711
Serial number (SN)	Unique item-level serial number	MH8031200000000001
Lot (DI = 1T)	Batch or lot number	110780
Quantity (DI = Q)	Quantity contained or represented	21
Country of Origin	ISO 3166-1 alpha-2 code	US (DI = 4L)

Component	Meaning	Value
Data carrier	2D code example	
NOTE This example is also a valid Identification Link (IL) string for products per EN IEC 61406-2.		

B.3.5 ID Scheme 5: Digital Object Identifier for products

See Table B.22 for an example of 5.6.2 using ISO 26324.

Table B.22 — Example of identification with DOI for products

Component	Meaning	Value
URL scheme	DOI resolver domain	https://doi.org
DOI Indicator	Constant prefix defined in ISO 26324 for DOI	10
Issued Org Code	Code assigned by the DOI issuing agency	1000
Product ID (Suffix)	Unique ID assigned by the org or agency (e.g. product, batch, item)	N44APB7E
Resolution	DOI resolver translates to the digital product passport location	Via doi.org
Data carrier	2D code example	
Full Identifier	ISO 26324 compliant identifier including prefix and suffix	10.1000/N44APB7E
Full example	https://doi.org/10.1000/N44APB7E	
NOTE This example is also a valid Identification Link (IL) string for products per EN IEC 61406-2.		

Annex C
(informative)

Overview of ID schemes for economic operators and facilities

C.1 General information for ID schemes

C.1.1 Governance and uniqueness

See Table C.1 for general information about governance and uniqueness on ID schemes for economic operators and facilities.

Table C.1 — Governance and uniqueness on ID schemes for economic operators and facilities

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
Identifier standard	ISO/IEC 6523.	ISO/IEC 15418.	DIDs v1.0:2022.	ISO 26324.
Type of issuing	Issuing agency.	Issuing agency.	Self-issuing system.	Issuing agency.
Unique economic operator identifier	Yes.	Yes.	Yes.	Yes.
Unique facility identifier	Yes.	Yes.	Yes.	No.
Uniqueness	ISO/IEC 6523-1.	ISO/IEC 15459 or EN IEC 61406-2.	DIDs v1.0:2022.	ISO 26324.
Reliance of domain name (for the unique economic operator and facility)	No, unique identifier is independent of domain name.	No, unique identifier is independent of domain name.	No, identifier is self-resolving using DID method syntax.	Yes.

C.1.2 Security

See Table C.2 for general information about security on ID schemes for economic operators and facilities.

Table C.2 — Security on ID schemes for economic operators and facilities

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
Can be combined with ISO/IEC 20248	Yes, supports digital signature for data authentication.		Yes, for when the DID is encoded in RAIN RFID (UHF RFID) per ISO/IEC 18000-63, or when a separate data carrier is used from a DID barcode.	Yes, for when the DOI is encoded in RAIN RFID (UHF RFID) per ISO/IEC 18000-63, or when a separate data carrier is used from a DOI barcode.
Can be combined with ISO 22376 [15]	Yes, compatible for data element structure and transfer syntax.	Yes, for ASC MH10.8.2 DI (structure + tagged data). No, for GS1 AIs.	No.	Yes, compatible for data element structure and transfer syntax.

C.1.3 Issuing process and prerequisites

See for Table C.3 general information about issuing process and prerequisites on ID schemes for economic operators and facilities.

Table C.3 — Issuing process and prerequisites on ID schemes for economic operators and facilities

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
Prerequisite for web enabled access to DPP	Can be embedded in URL.	Can be embedded in URL.	DID resolver and method required.	Resolvable via DOI resolver.

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Process of issuing of identifier	1. Get ICD and Org ID, 2. Issue identifiers.	1. Get CIN (ISO/IEC 15459 issuing agency) 2. Issue identifiers within assigned number band.	Self-issued using DID method.	1. Apply via DOI agency 2. Issue identifiers.
Sovereignty about the data in the product identifier	Economic operator controls ID elements.		Full sovereignty by economic operator.	Economic operator controls ID elements.
Changes in technology that would affect the product identifier	Changes in ISO/IEC standards or RFC 3986.			
Procedures for switching a service provider / issuing agency	No restrictions.			

C.2 Requirements and guidelines

C.2.1 Global uniqueness

See Table C.4 for information on requirements and guidelines of 4.1.

Table C.4 — Information on requirements and guidelines for global uniqueness

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
4.1.2 (1) No duplication	Ensured by the rules set in ISO/IEC 6523.	Ensured by the rules set in ISO/IEC 15459 or EN IEC 61406-2.	Yes.	Each ID is registered with unique metadata (checked by issuing agency).

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
4.1.2 (2) Distinct	Yes, ISO/IEC 6523-2.	Yes, globally unique and context independent per ISO/IEC 15459-3 or EN IEC 61406-2.	Yes, W3C DIDs v1.0:2022.	Yes, issuing agency checks that identifier is unique within prefix.
4.1.2 (3) Non-coexistence	Yes, prohibits reusing or reassigning identifiers.		Yes, prohibits reusing or reassigning identifiers ensured by cryptography and DID method rules.	Yes, prohibits reusing or reassigning identifiers.
4.1.2 (4) Cross domain identification	6.2	6.3	6.4	6.5

C.2.2 Persistence

See Table C.5 for information on requirements and guidelines of 4.2.

Table C.5 — Information on requirements and guidelines for persistence

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
4.2.2 (1) Consistency	Yes, per ISO/IEC 6523 and rules of listed ID schemes.	Yes, per ISO/IEC 15459-3, or EN IEC 61406-2 and the EU DPP system registry.	Yes, ensured by W3C DID v1.0 – persistence is technology-driven, not process-based.	Yes, ensured by requiring descriptive metadata for each entity provided to issuing agency.
4.2.2 (2) Preservation	Yes, per ISO/IEC 6523 and rules of listed ID schemes.	Yes, supported by life cycle management rules per ISO/IEC 15459-3 or per EN IEC 61406-1 and	Yes, through decentralized structure and governance throughout the life cycle.	Yes, issuing agency responsible for preservation of identifiers.

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
		−2 and the EU DPP system.		
4.2.2 (3) Permanence	Yes, per ISO/IEC 6523 and rules of listed ID schemes.	Yes, per ISO/IEC 15459-3 and issuing agency rules or per EN IEC 61406-1 and −2 and the EU DPP system.	Yes, under the control of the DID owner.	Yes, issuing agencies require the identifiers remain associated with the described referent.

C.2.3 Syntax

See Table C.6 for information on requirements and guidelines of 4.3.

Table C.6 — Information on requirements and guidelines for syntax

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
4.3.2 (1) Character set	ISO/IEC 646, IETF RFC 3986 and ISO/IEC 10646.			ISO 26324.
4.3.3 (1) Structure	Predefined (ICD + Org ID).	Predefined structure.	Partly fixed (did:<method>:) .	Prefix/Suffix structure.
4.3.3 (2) Variable length	Yes, variable within limits.	Yes, variable.		Variable suffix.

C.2.4 Granularity

See Table C.7 for information on requirements and guidelines of 4.4.

Table C.7 — Information on requirements and guidelines for granularity

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
4.4.3 (2) Resolver	Optional.	Optional.	Yes (DID resolver).	Yes (via doi.org).
4.4.3 (3) Hierarchy	Yes, depending on ID scheme.	Yes, possible to include.	Yes.	Yes, per issuing agency.

C.2.5 Interoperability

See Table C.8 for information on requirements and guidelines of 4.5.

Table C.8 — Information on requirements and guidelines for interoperability

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
4.5.2 (1) Data carrier	Web-enabled data carriers.			
4.5.2 (2) Digital access	Yes.			
4.5.2 (3) Compatibility	Yes, can be used with web technologies and existing supply chain equipment.	Yes, can be used with web technologies and existing supply chain equipment as per ISO/IEC 15459.	Yes, can be used with web technologies and existing supply chain equipment.	Yes, can be used with web technologies and existing supply chain equipment as per ISO/IEC 15459.
4.5.2 (5) Existing ID	Yes.			
4.5.3 (2) Portability	Yes, globally unique, cross-platform.			

C.2.6 Openness

See Table C.9 for information on requirements and guidelines of 4.6.

Table C.9 — Information on requirements and guidelines for openness

ID scheme	ID scheme 6	ID scheme 7	ID scheme 8	ID scheme 9
Reference	6.2	6.3	6.4	6.5
Title	Economic operator identification using ISO/IEC 6523	Economic operator and facility identification per ISO/IEC 15418	Economic operator identification in data systems using Decentralized identifiers (DID)	Economic operator identification used in data systems per Digital Object Identifiers (DOI)
4.6.2 (1) Transparency	Public ISO/IEC and web standards.	Public ISO/IEC and web standards.	Public W3C standards.	Public ISO/IEC and web standards.
4.6.2 (2) Consumer usage	No additional software needed.			
4.6.2 (3) No undue restrictions	No undue restrictions.		No undue restrictions, self-sovereign	No undue restrictions.

C.3 Examples

C.3.1 ID scheme 6: Economic operator identification schemes used in data systems per ISO/IEC 6523

See Table C.10 for examples of 6.2.2 ID schemes for organization included in ISO/IEC 6523-1. The examples are not all inclusive.

Table C.10 — Economic operator identification schemes used in data systems per ISO/IEC 6523

Name of ID scheme	ICD Code	Description
SIRENE	0002	The SIRENE number is used in France mainly for the official registration.
GS1 Party Global Location Number (Party GLN)	0088	Used to identify legal entities and, functional entities globally.
Dun and Bradstreet (DUNS)	0060	Unique business identifiers used globally for legal entities and company verification.
EU VAT	0223	Country specific VAT number.
Legal Entity Identifier (LEI)	0199	Identifies legal entities involved in financial transactions. Used for regulatory reporting, finance, and in supply chain transparency.

C.3.2 ID scheme 7: Economic operator and facility identifiers used in data carriers and data systems

See Table C.11 for examples of 6.3.2 using ASC MH10.8.2 DIs or GS1 AIs. The examples are not all inclusive.

Table C.11 — Examples of approaches using ASC MH10.8.2 DIs or GS1 AIs

Entity with example	AI or DI value	Description
ASC MH.10.8.2 Data Identifier for a party	DI (1V)	Supplier Code Assigned by Supplier.
ASC MH.10.8.2 Data Identifier for a party, conform to ISO/IEC 15459	DI (18V)	Identification of a party to a transaction in which the data format consists of two concatenated segments. The first segment is the Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, the second segment is a unique entity identification Company Identification Number (CIN) assigned in accordance with rules established by the issuing agency [16].
ASC MH.10.8.2 Data Identifier for a location	DI (1L)	Location.
ASC MH.10.8.2 Data Identifier for a location by its coordinates	DI (11L)	Location (Latitude/Longitude/Altitude) encoded in the format xnn.nnnnn/xnnn.nnnnnn/xnnnn5.
ASC MH.10.8.2 Data Identifier for a party by its VAT number	DI (24V)	Government-assigned Value Added Tax identification number identifying customer, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number.
GS1 Party - Global Location Number (Party GLN)	AI (417)	Used to identify legal entities and functional entities globally.
GS1 Global Location Number (GLN) for a physical location	AI (414)	Used to identify physical locations globally.

C.3.3 ID Scheme 8: Economic operator identification in data systems using Decentralized identifiers (DID)

See Table C.12 for an example of Decentralized identifiers (DIDs) for organizations or facilities.

Table C.12 — Example of economic operator or facility identification in data systems using Decentralized identifiers (DID)

Component	Meaning	Value
DID Scheme	DID method prefix as per DIDs v1.0:2022	did:
Method	DID method (e.g. web, ethr, ebsi)	web
Domain	Fully qualified domain controlled by the economic operator	example.com
Metadata	DID Document may include public keys, service end points, facility roles, granularity, etc.	Stored in DID Document

Component	Meaning	Value
Full example	did:web:example.com	

C.3.4 ID Scheme 9: Economic operator identification used in data systems per Digital Object Identifiers (DOI)

See Table C.13 for an example of 6.5.2 using ISO 26324.

Table C.13 — Example of economic operator identification used in data systems per Digital Object Identifiers (DOI)

Component	Meaning	Value
Scheme and Domain	Protocol and resolver domain	https://doi.org
Prefix (10.23)	Issuing agency prefix assigned under ISO 26324	10.23
Suffix	Identifier created by the economic operator or issuing agency	BB55–375D-4B8F-A0DD-3FEB
Full DOI	Complete identifier (prefix + suffix)	10.23/BB55-375D-4B8F-A0DD-3FEB
Resolution	Handled by DOI resolver, redirects to digital product passport URL	Redirect managed by doi.org
Full example	https://doi.org/10.23/BB55-375D-4B8F-A0DD-3FEB	

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of (EU) 2024/1781 aimed to be covered

This European Standard has been prepared under a Commission's standardization request M/604 AMD 1 to provide one voluntary means of conforming to essential requirements of (EU) Regulation of 2024/1781 of 13 June 2024 of the European Parliament and of the Council of the European Union establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC.

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding essential requirements of that Regulation, and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Articles of (EU) Regulation 2024/1781

Essential requirements specified in Article 10 and 11 of 2024/1781	Subclause(s) of this EN	Remarks/Notes
Article 10 1.(a)	4.1.2, 4.2.2, 4.5.2, 5.1	
Article 10 1.(b)	-	Not Applicable
Article 10 1.(c)	5.1, 3.1.25	
Article 10 1.(d)	4.3.2, 4.4.2, 4.5.2, 4.6.2	
Article 10 1.(e)	-	Not Applicable
Article 10 1.(f)	4.4.2	
Article 10 1.(g)	-	Not Applicable
Article 10 2	-	Not Applicable
Article 10 3.(a)	4.5.2	
Article 10 3.(b)	-	Not Applicable
Article 10 4	-	Not Applicable
Article 11 (a)	4.5.2, 4.6.2	
Article 11 (b)	4.6.2	
Article 11 (c)	-	Not Applicable
Article 11 (d)	4.4.2	
Article 11 (e)	4.2.2	Only applies to the persistence of the unique identifier.

Essential requirements specified in Article 10 and 11 of 2024/1781	Subclause(s) of this EN	Remarks/Notes
Article 11 (f)	-	Not Applicable
Article 11 (g)	-	Not Applicable
Article 11 (h)	-	Not Applicable
Article 12 (1)	6.1	
Article 12 (2)	6.1	
Article 12 (3)	6.1	
Article 12 (4)	-	Not Applicable
Article 12 (5)	-	Not Applicable
Article 12 (6)	-	Not Applicable

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the products falling within the scope of this standard.

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